Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 105305 | *Subject: Operating System* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only) **[2 x 7 = 14]**

(a) What is the ready state of a process?

(i) When process waits for the scheduler to do a context switch

(ii) When process is using the CPU

(iii) When process is unable to run until some task has been completed

(iv) None of the mentioned

(b) The total number of processes completed per unit time is known as _____

(i) Output

(ii) Throughput

(iii) Efficiency

(iv) Capacity

(c) Physical memory is broken into fixed–sized blocks called _____

(i) Frames

(ii) Backing store

(iii) Pages

(iv) None

(d) The First Fit, Best Fit and Worst Fit are partitioning algorithms to select a ____

(i) Process from a queue to put in memory

(ii) Free Hole from a set of available Hole

(iii) Processor to run the next process

(iv) All of the mentioned

(e) In Paging, every logical address generated by the CPU is divided into two parts. They are _____

(i) Page Number and Frame Bit

(ii) Page Offset and Frame Bit

(iii) Page Number and Page Offset

(iv) Frame Offset and Page Offset

(f) Page fault occurs when

(i) Page is not in cache

(ii) CPU idle

(iii) Page not in main memory

(iv) Page is modified

(g) Deadlock prevention ensures

(i) Deadlock never occurs

(ii) Deadlock recovery is easy

(iii) Deadlock is detected after occurrence

(iv) System ignores deadlocks

(h) Which IPC mechanism requires synchronization between processes?

(i) Shared memory

(ii) Pipes

(iii) Message passing

(iv) Signals

(i) User-level threads are managed by

(i) Kernel

(ii) Hardware

(iii) Thread library

(iv) Compiler

(j) A process moves from Running to Ready state due to

(i) I/O request

(ii) Time quantum expiry

(iii) Process termination

(iv) Waiting for signal

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Q.2 (a) What is dispatch latency? How does it affect Real time scheduling? Suggest some solutions to keep dispatch latency low. [7]

(b) What is a process scheduler? State the characteristics of an exemplary process scheduler. [7]

Q.3 (a) Describe the implementation of semaphores in attaining process synchronization. Explain any two classical process synchronization problems. [7]

(b) Consider a disk with 200 tracks and the queue has random requests from different processes in the order: 55, 58, 39, 18, 90, 160, 150, 38, 184. Initially arm is at 100. Find the Average Seek length using FIFO, SSTF and SCAN algorithm. [7]

Q.4 (a) Consider the set of Processes < A, B, C, D > assumed to have arrived at the time sequence <0, 1, 2, 4> having Burst time < 3, 6, 4, 2> respectively. Draw the Giant Chart illustrating their execution using Shortest Job First (SJF) Algorithm, Shortest Remaining Time First (SJF Preemptive Algorithm) and calculate the Average Turn Around time and Average waiting time for both scheduling algorithms. [7]

(b) With a neat sketch (process state diagram), explain various states the process may undergo from its creation to its termination. Mark all possible transitions and states and clearly explain the details of states and state-transitions. [7]

Q.5 (a) Consider the following snapshot of a system: [7]

Allocation	A	B	C	D	Max: A	B	C	D
P0	0	0	1	2	0	0	1	2
P1	1	0	0	0	1	5	2	0
P2	1	3	5	4	1	7	5	0
P3	0	6	3	2	2	3	5	6
P4	0	0	1	4	0	6	5	2
Available	A	B	C	D				
	0	6	5	6				

Answer the following questions using the banker's algorithm: (i) What is the content of the matrix Need? (ii) Is the system in a safe state? (iii) If a request from process P1 arrives for (0,4,2,0), can the request be granted immediately?

(b) Explain the concept of paging in operating systems. Discuss the structure of logical and physical addresses, the role of the page table in address translation, and analyze the issue of internal fragmentation in paging systems. [7]

Q.6 (a) Consider the following page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. How many page fault would occur for the following page replacement algorithms for 3 page frame? All frames are initially empty. [7]
(i) FIFO replacement
(ii) LRU
(iii) Optimal replacement

(b) Describe the segmentation memory management scheme in operating systems. Explain the logical address format, segment table structure, and discuss the problem of external fragmentation along with its impact on memory utilization. [7]

Q.7 (a) Explain the concept of preemptive CPU scheduling in operating systems. Discuss its working mechanism, advantages, disadvantages, and compare it with non-preemptive scheduling in terms of responsiveness and overhead. [7]

(b) Discuss the significance of time quantum in time-sharing systems. Analyze how the choice of time slice affects CPU utilization, context switching overhead, and turnaround time of processes. [7]

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Q.8 (a) Explain the concept of multithreading in operating systems. Discuss the advantages of multithreading, different models of multithreading, and analyze the challenges involved in thread management such as synchronization and resource sharing. [7]

(b) In a system implementing the Readers–Writers synchronization problem, suppose there are 5 processes arriving in the following order with their arrival times (in ms) and operation durations: [7]

(i) R1: Arrival = 0, Duration = 4
(ii) W1: Arrival = 1, Duration = 3
(iii) R2: Arrival = 2, Duration = 2
(iv) R3: Arrival = 3, Duration = 3
(v) W2: Arrival = 5, Duration = 2

Assuming the first Readers–Writers problem (reader-priority) is followed: Determine the execution order of processes and compute the waiting time for each process. Also, draw the corresponding timeline (Gantt chart) showing access to the shared resource.

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Batch Procession System
(b) Multiprogramming System
(c) Real-Time System
(d) Timesharing System

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